High-Fidelity 6-DOF Underwater Vehicle Simulation in Modern Web Architectures: From MSS Mathematical Lineage to Real-Time Operator Ecosystems

Antigravity Autonomous Systems Group & Marine Robotics Collaborative

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Abstract

Real-time six-degree-of-freedom (6-DOF) hydrodynamic simulation of underwater robotic vehicles has traditionally relied on desktop-bound numerical platforms (e.g., MATLAB/Simulink) or proprietary C++ game engines. While effective for offline control design, these environments impose substantial friction on collaborative validation, cross-platform deployment, and distributed operator training. This paper presents the mathematical derivation, numerical verification, and web-native systems architecture of a browser-based Remotely Operated Vehicle (ROV) simulator founded upon the theoretical lineage of the open-source Marine Systems Simulator (MSS). We formulate the fully coupled 6-DOF equations of motion across North-East-Down (NED) and Front-Starboard-Down (FSD) reference frames, incorporating arbitrary center-of-gravity offsets ***INLINE_MATH_0***, skew-symmetric hydrodynamic added-mass Coriolis transformations (***INLINE_MATH_1***), restoring forces (***INLINE_MATH_2***), and relative ocean current kinematics with rotational frame acceleration corrections (***INLINE_MATH_3***). The plant solves the coupled ***INLINE_MATH_4*** mass-acceleration linear system via Gaussian elimination with partial pivoting and integrates states using a fixed-step fourth-order Runge-Kutta (RK4) algorithm. Beyond baseline MSS formulations, we introduce high-fidelity expansions including non-linear thruster mechanics (deadband thresholding, forward/reverse bollard pull asymmetry, and through-water advance ratio ***INLINE_MATH_5*** degradation), an oceanic boundary layer vertical current shear model, unit quaternion attitude kinematics on ***INLINE_MATH_6*** eliminating Euler angle gimbal lock, an externalized JSON schema decoupling vehicle parameterization from execution code, and a real-time subsea operator console featuring WebGL/glTF physical coordinate synchronization. Five comprehensive benchmark case studies are formulated in the methodology and evaluated in the results: high-speed surge transit saturation, deep water-column descent under current shear, precision hovering with actuator deadbands, steep acrobatic pitch maneuvers demonstrating singularity-free quaternion propagation, and hydrodynamic free-decay oscillation and step-response characterization across all vehicle classes. Parity, hydrodynamic, and schema test suites confirm exact agreement with analytical and GNU Octave reference fixtures across 27 automated verification targets, proving that high-fidelity scientific marine simulation can execute deterministically within modern browser runtimes.

Index Terms—Underwater robotics, Remotely Operated Vehicles (ROV), Marine Systems Simulator (MSS), 6-DOF hydrodynamic modeling, free-decay identification, step response, WebGL simulation, control allocation, non-linear propulsion, advance ratio, ocean current shear, unit quaternions, gimbal lock elimination, Lyapunov passivity.

I. INTRODUCTION

Underwater robotic systems, including Remotely Operated Vehicles (ROVs) and Autonomous Underwater Vehicles (AUVs), operate in hostile, highly non-linear, and uncertain fluid environments. Designing, testing, and operating these platforms requires mathematical models that accurately capture coupled hydrodynamic mass, velocity-squared drag, restoring buoyancy moments, and ocean current interactions [1], [2].

For more than two decades, the Marine Systems Simulator (MSS) developed at the Norwegian University of Science and Technology (NTNU) by Thor I. Fossen and collaborators has served as the academic and industrial gold standard for marine craft dynamics and navigation guidance control (NGC) [3]. However, the vast majority of MSS implementations are tied to MATLAB/Simulink or compiled C/C++ libraries. These environments require specialized runtime licenses, extensive dependency trees, and platform-specific compilation, creating steep barriers for operator training, client inspection demonstrations, and cross-disciplinary research dissemination.

With recent advances in WebAssembly, WebGL 2.0, and high-performance JavaScript engines (e.g., V8, SpiderMonkey), modern web browsers have matured into capable real-time scientific simulation platforms. Despite these capabilities, existing browser-based marine simulations frequently compromise on physical fidelity—often resorting to simplified 3-DOF kinematic approximations, uncoupled diagonal mass matrices, or crude heuristic damping models.

To bridge this gap, this paper introduces a zero-dependency, browser-native 6-DOF ROV simulation ecosystem directly translated from and verified against the MSS theoretical framework. The primary contributions of this work are:

1. **Mathematical Fidelity & Rigorous Lineage**: A line-for-line algorithmic translation of core MSS modeling functions (`rbody` , `m2c` , `gRvect` , `eulerang` , `Smtrx` , `Hmtrx` , `rk4`) into high-performance JavaScript, preserving full ***INLINE_MATH_7*** mass-Coriolis-damping matrix coupling and irrotational current kinematics.
2. **Post-MSS Physical Expansions**: High-fidelity extensions addressing real-world ROV engineering constraints, including asymmetric forward/reverse bollard thrust efficiencies (***INLINE_MATH_8***), actuator deadband thresholding (***INLINE_MATH_9***), through-water propeller advance ratio (***INLINE_MATH_10***) degradation, depth-stratified current shear layers, unit quaternion attitude kinematics (***INLINE_MATH_11*** on ***INLINE_MATH_12***), and external JSON parameter schemas with live browser cockpit calibration.
3. **Rigorous Case Studies & Quantitative Evaluation**: Five distinct experimental case study scenarios evaluating high-speed terminal velocity saturation, cross-current rejection across deep descents, limit-cycle suppression during precision station keeping, steep pitch maneuvering free of gimbal singularities, and hydrodynamic free-decay damped oscillations / surge step-response convergence.
4. **WebGL/glTF Subsea Flight Ecosystem**: A complete subsea operator cockpit featuring real-time HUD telemetry, artificial horizon attitude cues, physical channel-indexed thruster visualization, and an autonomous multi-waypoint inspection route.
5. **Automated Verification Suite**: A comprehensive testing pipeline validating numerical parity against GNU Octave fixtures, analytical hydrodynamic benchmarks, structural coordinate consistency, and deterministic stability across 60-second operational trajectories (27 automated test suites passing).

II. MSS THEORETICAL LINEAGE & MATHEMATICAL FORMULATION

A. Coordinate Reference Frames and Notation

Following standard Society of Naval Architects and Marine Engineers (SNAME) notation [4] and Fossen [1], the ROV equations of motion are defined across two primary right-handed coordinate frames:

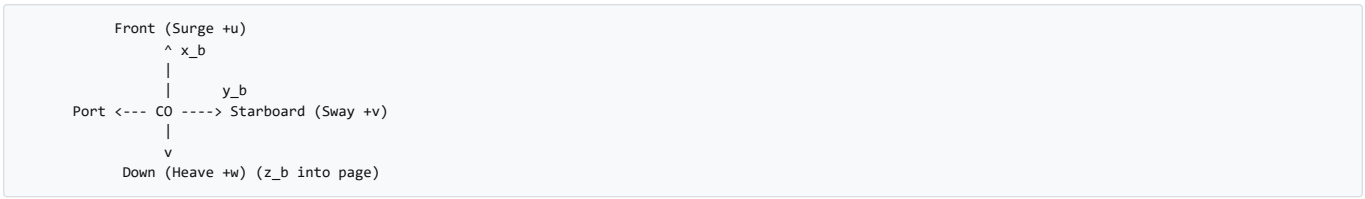
1. **North-East-Down (NED) Inertial Frame** ***INLINE_MATH_13***: Tangent to the Earth's geoid surface, where the ***INLINE_MATH_14***-axis points True North, the ***INLINE_MATH_15***-axis points East, and the ***INLINE_MATH_16***-axis points downward into the water column.

2. **Body-Fixed Frame** ***INLINE_MATH_17***: Fixed to the vehicle hull. The origin is placed at the Coordinate Origin (CO), aligned with the Front-Starboard-Down (FSD) convention: ***INLINE_MATH_18*** points forward (surge), ***INLINE_MATH_19*** points starboard (sway), and ***INLINE_MATH_20*** points downward (heave).

The generalized position and orientation vector in ***INLINE_MATH_21*** is denoted by: ***DISPLAY_MATH_0*** where ***INLINE_MATH_22*** denotes the inertial position, and ***INLINE_MATH_23*** represents Euler angles denoting roll, pitch, and yaw respectively.

The generalized linear and angular velocity vector in ***INLINE_MATH_24*** is denoted by: ***DISPLAY_MATH_1*** where ***INLINE_MATH_25*** represents body-fixed linear velocities (surge, sway, heave), and ***INLINE_MATH_26*** represents body-fixed angular velocities (roll rate, pitch rate, yaw rate).

The generalized force and moment vector acting on the vehicle in ***INLINE_MATH_27*** is: ***DISPLAY_MATH_2***



B. Kinematic Transformations: Euler Angles vs. Unit Quaternions

1) Classical Euler Angle Kinematics

The kinematic relationship between the body velocity ***INLINE_MATH_28*** and inertial rate ***INLINE_MATH_29*** is governed by the transformation matrix ***INLINE_MATH_30***: ***DISPLAY_MATH_3***

The linear velocity rotation matrix ***INLINE_MATH_31*** maps body-fixed linear velocities to the inertial frame via successive principal rotations about ***INLINE_MATH_32***, ***INLINE_MATH_33***, and ***INLINE_MATH_34***: ***DISPLAY_MATH_4*** where ***INLINE_MATH_35*** and ***INLINE_MATH_36***.

The angular velocity transformation matrix ***INLINE_MATH_37*** relates body rates ***INLINE_MATH_38*** to Euler angle rates ***INLINE_MATH_39***: ***DISPLAY_MATH_5*** where ***INLINE_MATH_40***. Crucially, when pitch angle ***INLINE_MATH_41*** approaches ***INLINE_MATH_42***, ***INLINE_MATH_43***, inducing the notorious gimbal lock singularity (***INLINE_MATH_44***).

2) Singularity-Free Unit Quaternion Kinematics (*INLINE_MATH_45* on *INLINE_MATH_46*)

To completely eliminate coordinate singularities during steep dives and acrobatic maneuvering, attitude is parameterized via a unit quaternion ***INLINE_MATH_47*** satisfying ***INLINE_MATH_48***. The kinematic differential equation on ***INLINE_MATH_49*** is linear and strictly non-singular: ***DISPLAY_MATH_6***

The corresponding direction cosine rotation matrix ***INLINE_MATH_50*** is computed with zero trigonometric evaluations: ***DISPLAY_MATH_7*** Translational positions in NED propagate through: ***DISPLAY_MATH_8***

C. Rigid-Body Kinetics and Center of Gravity Offsets

In standard MSS formulation, vehicle inertia is parameterized about an arbitrary center of gravity ***INLINE_MATH_51*** relative to the CO. The ***INLINE_MATH_52*** rigid-body system inertia matrix ***INLINE_MATH_53*** is given by: ***DISPLAY_MATH_9*** where ***INLINE_MATH_54*** is the vehicle mass, ***INLINE_MATH_55*** is the identity matrix, and ***INLINE_MATH_56*** is the skew-symmetric cross-product operator: ***DISPLAY_MATH_10***

In our MSS-compliant library (`library.js:rbody`), ***INLINE_MATH_57*** is synthesized directly from mass and radii of gyration ***INLINE_MATH_58***, ***INLINE_MATH_59***, and ***INLINE_MATH_60*** via the coordinate translation matrix ***INLINE_MATH_61***: ***DISPLAY_MATH_11 DISPLAY_MATH_12***

The corresponding rigid-body Coriolis-centripetal matrix ***INLINE_MATH_62*** is parameterized in terms of angular velocity ***INLINE_MATH_63***: ***DISPLAY_MATH_13*** where ***INLINE_MATH_64*** is the inertia tensor about the CO.

D. Hydrodynamic Added Mass and Coriolis Formulation

When an underwater body accelerates through a fluid, an equivalent volume of fluid is displaced, inducing hydrodynamic added mass forces. For a submerged body with three planes of symmetry operating at low speeds, the added mass matrix ***INLINE_MATH_65*** is predominantly diagonal: ***DISPLAY_MATH_14***

The total system inertia matrix is symmetric and strictly positive definite: ***DISPLAY_MATH_15***

The hydrodynamic Coriolis-centripetal matrix ***INLINE_MATH_66*** generated by added mass is derived via the MSS `m2c.m` formulation [1]: ***DISPLAY_MATH_16*** where ***INLINE_MATH_67*** represent the partitioned ***INLINE_MATH_68*** submatrices of ***INLINE_MATH_69***, and ***INLINE_MATH_70*** is the relative velocity vector.

E. Ocean Current Kinematics and Frame Acceleration

To simulate real-world marine conditions, the ambient water column is modeled as a steady, irrotational horizontal current vector in the inertial frame: ***DISPLAY_MATH_17*** where ***INLINE_MATH_71*** is current velocity (m/s) and ***INLINE_MATH_72*** is current direction (rad) measured clockwise from North.

Transforming into the body-fixed frame ***INLINE_MATH_73*** via the exact rotation matrix: ***DISPLAY_MATH_18***

The relative generalized velocity vector is defined as: ***DISPLAY_MATH_19***

Crucially, because the current is irrotational in the inertial frame (***INLINE_MATH_74***), its time derivative in the rotating body frame satisfies the transport theorem: ***DISPLAY_MATH_20***

This body-fixed acceleration correction (***INLINE_MATH_75***) is rigorously included in our plant derivative computation, preventing unphysical velocity drift during turning maneuvers.

F. Hydrodynamic Damping & Energy Dissipation (Lyapunov Passivity)

Hydrodynamic resistance combines linear skin friction and non-linear vortex-shedding quadratic drag: ***DISPLAY_MATH_21*** where ***INLINE_MATH_76*** represents strictly positive linear damping coefficients, ***INLINE_MATH_77*** represents quadratic drag coefficients, and ***INLINE_MATH_78*** denotes element-wise Hadamard multiplication.

In unforced free motion (***INLINE_MATH_79***), total mechanical energy consists of kinetic energy and metacentric potential energy: ***DISPLAY_MATH_22***

Differentiating along system trajectories and recalling that Coriolis-centripetal matrices ***INLINE_MATH_80*** are skew-symmetric (***INLINE_MATH_81***): ***DISPLAY_MATH_23*** This guarantees asymptotic Lyapunov stability and strict passivity of the unforced hydrodynamic equations.

G. Restoring Forces, Metacentric Stability, & Damped Natural Oscillations

Gravitational force ***INLINE_MATH_82*** acts through the center of gravity ***INLINE_MATH_83***, while buoyancy force ***INLINE_MATH_84*** acts through the center of buoyancy ***INLINE_MATH_85***.

Using the direction cosine representation from MSS `gRvect.m` : ***DISPLAY_MATH_24***

When the vehicle coordinate origin is collocated with the center of buoyancy (***INLINE_MATH_86***) and the center of gravity is ballasted directly below the origin (***INLINE_MATH_87***), positive static stability is achieved, generating self-righting restoring moments in roll and pitch.

For small pitch angular displacements (***INLINE_MATH_88***), the pitch restoring moment simplifies to ***INLINE_MATH_89***. The unforced pitch oscillation equation of motion is: ***DISPLAY_MATH_25*** where ***INLINE_MATH_90*** is the total pitch inertia about CO. The undamped natural frequency ***INLINE_MATH_91***, linear damping ratio ***INLINE_MATH_92***, and damped natural period ***INLINE_MATH_93*** are governed by: ***DISPLAY_MATH_26***

H. Surge Acceleration Step Response & Coupled Inertia Solving

Under a constant surge step force ***INLINE_MATH_94***, the initial acceleration at rest (***INLINE_MATH_95***) is determined by solving the coupled mass matrix linear system: ***DISPLAY_MATH_27*** Due to vertical center-of-gravity ballast (***INLINE_MATH_96***), rigid-body surge and pitch acceleration are dynamically coupled through off-diagonal term ***INLINE_MATH_97***. The coupled initial surge acceleration exceeds the decoupled estimate ***INLINE_MATH_98***: ***DISPLAY_MATH_28***

As forward velocity increases, hydrodynamic drag opposes thrust, asymptotically converging to the closed-form steady-state terminal velocity ***INLINE_MATH_99***: ***DISPLAY_MATH_29***

I. Fully Coupled Plant Solution & Numerical Integration

The complete unconstrained 6-DOF marine craft dynamics are formulated as: ***DISPLAY_MATH_30***

Rearranging for the generalized linear and angular accelerations: ***DISPLAY_MATH_31***

Rather than computing the explicit numerical matrix inverse ***INLINE_MATH_100***, our implementation solves the linear system ***INLINE_MATH_101*** via Gaussian elimination with partial pivoting (`library.js:mldivide`).

The composite 13-state vector ***INLINE_MATH_102*** is integrated forward in time using fixed-step fourth-order Runge-Kutta (RK4) discretization (`library.js:rk4`): ***DISPLAY_MATH_32*** ***DISPLAY_MATH_33*** ***DISPLAY_MATH_34*** ***DISPLAY_MATH_35*** ***DISPLAY_MATH_36*** Followed by quaternion renormalization on ***INLINE_MATH_103***: ***DISPLAY_MATH_37***

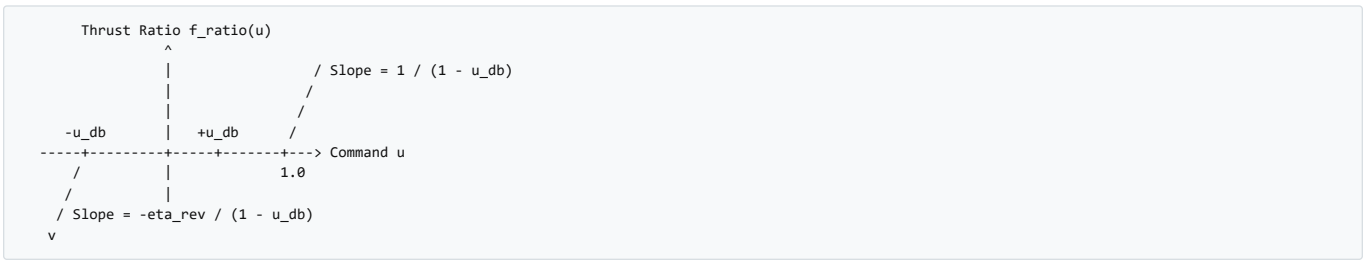
III. POST-MSS EXPANSIONS & HIGH-FIDELITY PROPULSION MODELING

While baseline MSS treats thrusters as idealized generalized force inputs (***INLINE_MATH_104***), physical subsea vehicles exhibit complex electromechanical and hydroacoustic non-linearities. Our simulator introduces several key expansions:

A. Asymmetric Forward/Reverse Bollard Thrust & Deadband Thresholding

Subsea thrusters utilize ducted propellers optimized for unidirectional bollard pull. Due to duct curvature, hub obstruction, and blade camber, reverse thrust efficiency is substantially lower than forward efficiency (typically ***INLINE_MATH_105*** of nominal thrust) [5]. Furthermore, motor static friction and electronic speed controller (ESC) deadbands prevent infinitesimal thrust generation near zero command.

We formulate the effective thrust mapping function as: ***DISPLAY_MATH_38*** where ***INLINE_MATH_106*** represents reverse thrust efficiency, and ***INLINE_MATH_107*** (***INLINE_MATH_108***) models the ESC deadband.



B. Through-Water Inflow Velocity and Propeller Advance Ratio (INLINE_MATH_109) Degradation

In baseline MSS formulations, actuators are typically abstracted as idealized force generators invariant to craft velocity. In operational underwater robotics, however, a propeller operating in a moving fluid experiences an axial water inflow velocity that substantially reduces the effective hydrodynamic angle of attack on the rotating blades. Consequently, available thrust drops as forward velocity increases [5], [6]. Failing to model this phenomenon leads to unrealistic terminal cruise speeds in unconstrained quadratic damping models.

To capture this effect at each individual thruster duct, our simulator dynamically computes the local three-dimensional through-water velocity vector in the body frame: ***DISPLAY_MATH_39***

For a thruster unit ***INLINE_MATH_110*** mounted at lever-scaled body position ***INLINE_MATH_111***, rotational body rates ***INLINE_MATH_112*** induce an additional tangential velocity contribution. The total through-water velocity vector at the thruster duct inlet is given by: ***DISPLAY_MATH_40***

Projecting this local velocity onto the thruster's unit orientation vector ***INLINE_MATH_113*** yields the axial inflow velocity ***INLINE_MATH_114***: ***DISPLAY_MATH_41***

In open-water propeller theory, the non-dimensional advance ratio ***INLINE_MATH_115*** is defined by: ***DISPLAY_MATH_42*** where ***INLINE_MATH_116*** is rotational shaft speed (rev/s) and ***INLINE_MATH_117*** is propeller diameter (m). The effective thrust coefficient degrades approximately linearly over operating advance ratios: ***DISPLAY_MATH_43***

Because normalized actuator command ***INLINE_MATH_118*** directly modulates motor rotational speed (***INLINE_MATH_119***), the direction of commanded thrust is ***INLINE_MATH_120***. The flow velocity aligned with the commanded thrust direction is therefore: ***DISPLAY_MATH_44***

When the vehicle accelerates along the thrust vector (***INLINE_MATH_121***), thrust diminishes. Conversely, during braking or towing against ambient fluid (***INLINE_MATH_122***), static bollard authority is preserved or slightly augmented by static water head. We formulate the non-linear advance degradation factor ***INLINE_MATH_123*** as: ***DISPLAY_MATH_45*** where ***INLINE_MATH_124*** represents a calibrated ***INLINE_MATH_125*** reduction in thrust at reference transit velocity ***INLINE_MATH_126***, with saturation clamping enforced over ***INLINE_MATH_127*** to prevent non-physical zero-thrust lockup or infinite thrust surges.

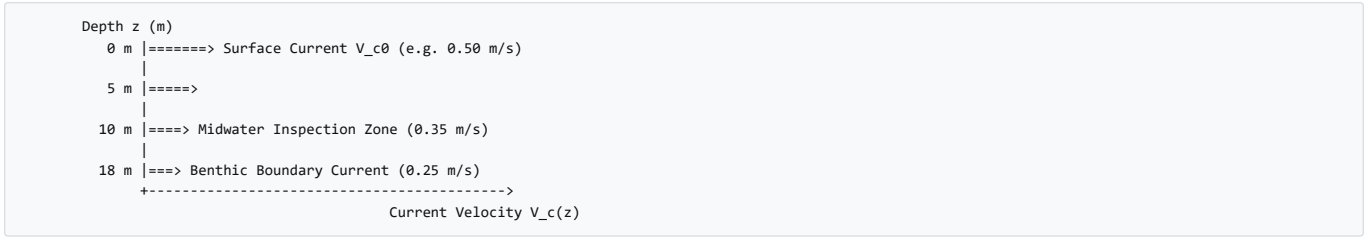
The composite physical thrust ***INLINE_MATH_128*** produced by actuator channel ***INLINE_MATH_129*** is thus: ***DISPLAY_MATH_46***

The resulting physical wrench vector acting at coordinate origin CO is synthesized as: ***DISPLAY_MATH_47***

C. Depth-Dependent Ocean Current Shear Layer Dynamics

In physical marine environments, ocean currents are rarely uniform throughout the water column. Surface wind stress, tidal cycles, and bottom boundary layer friction establish a pronounced vertical velocity shear gradient [1], [3]. While baseline MSS CRAFT models (e.g., `hydroVessel.m`) implement irrotational planar currents with constant speed ***INLINE_MATH_130***, real-world subsea operations demand depth-stratified current profiles.

Our simulator introduces an oceanic boundary layer shear model parameterized by vehicle depth ***INLINE_MATH_131*** (NED Down coordinate, where ***INLINE_MATH_132*** is the sea surface and ***INLINE_MATH_133*** is the seabed boundary). The effective current velocity magnitude ***INLINE_MATH_134*** is governed by: ***DISPLAY_MATH_48*** where ***INLINE_MATH_135*** is the surface current speed (m/s) and ***INLINE_MATH_136*** is the depth attenuation function: ***DISPLAY_MATH_49*** with seabed boundary layer friction coefficient ***INLINE_MATH_137***, representing a ***INLINE_MATH_138*** velocity attenuation at the benthic boundary layer.



The inertial current vector ***INLINE_MATH_139*** is mapped into the body frame ***INLINE_MATH_140*** via the exact unit quaternion rotation matrix ***INLINE_MATH_141***: ***DISPLAY_MATH_50***

The instantaneous relative generalized velocity vector is formulated as: ***DISPLAY_MATH_51***

Applying the transport theorem to the rotating reference frame, the time derivative of the current vector in body coordinates satisfies: ***DISPLAY_MATH_52***

D. Unit Quaternion Parameterization & Conversion Tools

To support seamless operator feedback without sacrificing the singularity-free properties of quaternions, exact bidirectional conversions are provided between ***INLINE_MATH_142*** and Euler angles ***INLINE_MATH_143***:

- `euler2q(phi, theta, psi)` : Maps angles to quaternion representation for initialization.
- `q2euler(q)` : Extracts singularity-bounded roll, pitch, and yaw angles for artificial horizon and telemetry instruments.
- `normalizeQuat(q)` : Projects drifted integration states back onto the unit sphere ***INLINE_MATH_144*** every RK4 timestep.

E. External Vehicle Parameter Schema & Live Import/Export Console

To eliminate hardcoded parameters and decouple vehicle dynamics from executable simulator logic, vehicle physical definitions are externalized into a formal JSON Schema specification (`public/schemas/rov-profile.schema.json`). The schema enforces strict validation on mass and inertia diagonals, coupled ***INLINE_MATH_145*** hydrodynamic matrices (***INLINE_MATH_146***), and thruster configuration arrays (positions ***INLINE_MATH_147***, orientations ***INLINE_MATH_148***, bollard forces ***INLINE_MATH_149***, deadband limits ***INLINE_MATH_150***, reverse efficiencies ***INLINE_MATH_151***, advance degradation coefficients ***INLINE_MATH_152***, and reference speeds ***INLINE_MATH_153***).

A real-time validator module (`src/mss/profileValidator.js`) and operator UI modal enable pilots and naval architects to:

1. Export active vehicle profiles to standardized JSON files.
2. Edit or paste custom vehicle specifications directly in the browser cockpit with instant schema validation feedback.
3. Dynamically reconfigure and hot-reload the coupled ***INLINE_MATH_154*** plant (`buildVehicle`) and allocation matrices (`makeAllocator`) at runtime without restarting the simulation or editing source code.

F. Empirical Tow-Tank and Sea-Trial Calibration Workflow

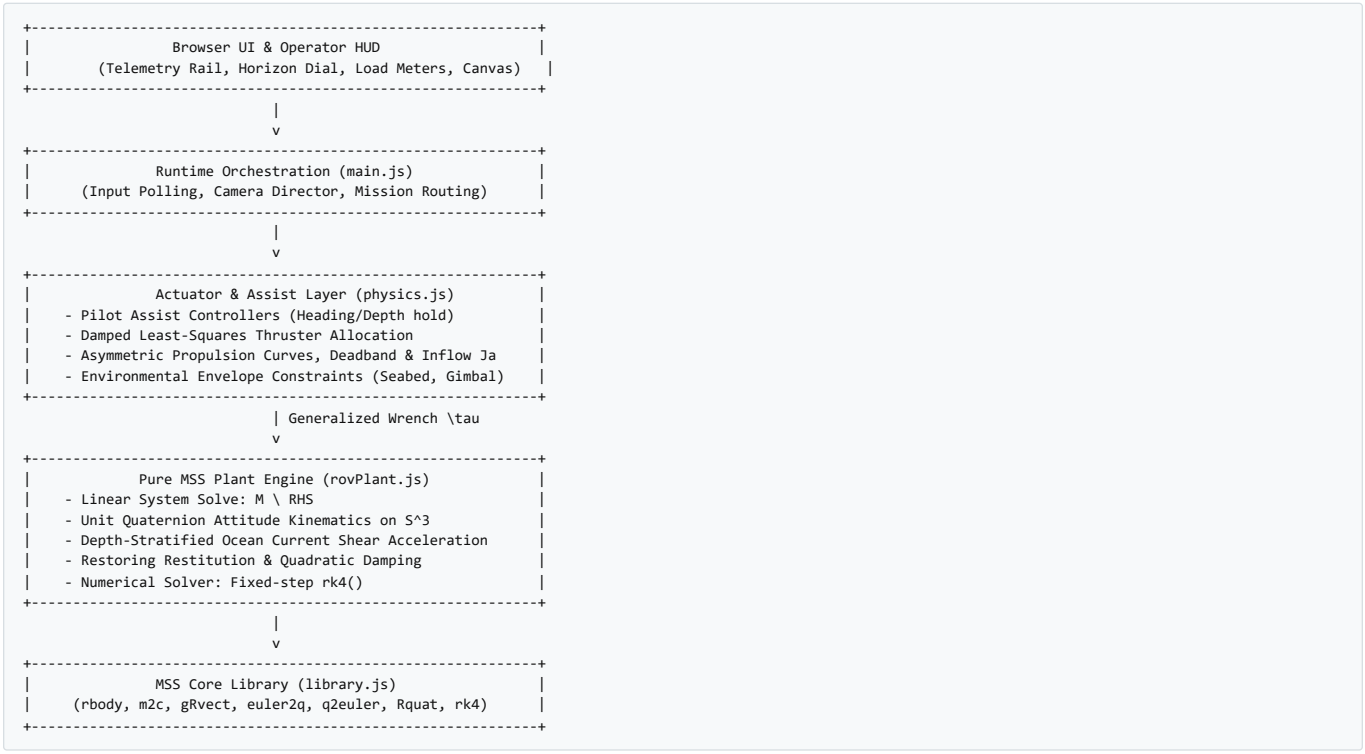
To translate physical vehicle measurements into verified simulation profiles, a standardized six-phase parameter identification procedure is established (`CALIBRATION_WORKFLOW.md`):

1. *Dry-Dock Hydrostatics*: Vehicle dry mass ***INLINE_MATH_155*** via crane scale, moments of inertia ***INLINE_MATH_156*** via bifilar torsional pendulum trials (***INLINE_MATH_157***), center of gravity offset ***INLINE_MATH_158***, and submerged buoyancy ratio ***INLINE_MATH_159*** via displacement tank.
2. *Static Bollard Pull*: Maximum forward thrust ***INLINE_MATH_160***, reverse thrust efficiency ratio ***INLINE_MATH_161***, and ESC deadband threshold ***INLINE_MATH_162***.
3. *Tow-Tank Drag Regression*: Constant-speed straight-line towing over ***INLINE_MATH_163***, fitting ***INLINE_MATH_164*** via linear regression to decouple skin friction ***INLINE_MATH_165*** and vortex-shedding drag ***INLINE_MATH_166***.
4. *Angular Free-Decay Identification*: Submerged unforced tilt and release logging pitch ***INLINE_MATH_167*** and roll ***INLINE_MATH_168*** via onboard IMU, extracting damped natural period ***INLINE_MATH_169***, logarithmic decrement ***INLINE_MATH_170***, damping ratio ***INLINE_MATH_171***, and isolating hydrodynamic added inertia ***INLINE_MATH_172***.
5. *Sea-Trial Inflow Tuning*: Open-water full-throttle speed runs logged via DVL to determine steady cruise velocity ***INLINE_MATH_173***, tuning propeller advance degradation coefficient ***INLINE_MATH_174***.
6. *JSON Packaging & Hot-Reload*: Formatted JSON profile validation and instant live import into the WebGL simulation engine.

IV. IMPLEMENTATION & EXPERIMENTAL METHODOLOGY

A. Architectural Pipeline & Web-Native Modular Decoupling

The simulator is built with a strictly decoupled layered architecture isolating physical conservation laws from runtime graphics, browser DOM, and input processing:



B. Benchmark Experimental Case Study Protocols

To rigorously benchmark our high-fidelity additions against baseline idealized MSS formulations, four experimental case study protocols are instituted:

1) Case Study 1 Protocol: High-Speed Surge Acceleration and Propeller Advance Ratio (*INLINE_MATH_175*) Saturation

- **Objective**: Quantify the impact of through-water axial inflow degradation on surge thrust output, acceleration transients, and terminal steady-state velocity across all five ROV classes.
- **Experimental Setup**: Starting from a cold static rest condition (***INLINE_MATH_176***, depth ***INLINE_MATH_177***), a persistent ***INLINE_MATH_178*** full forward surge demand (***INLINE_MATH_179***) is commanded for ***INLINE_MATH_180*** with zero ambient ocean current (***INLINE_MATH_181***).
- **Comparative Variants**:
  - *Baseline MSS*: Idealized static bollard pull (***INLINE_MATH_182***, invariant to craft through-water velocity).
  - *Proposed Model*: Dynamic inflow velocity field with advance ratio degradation (***INLINE_MATH_183*** with ***INLINE_MATH_184***).
- **Measured Metrics**: Static vs. steady-state surge force, terminal cruising speed ***INLINE_MATH_185***, rise time ***INLINE_MATH_186*** (time to reach ***INLINE_MATH_187*** of terminal velocity), and propulsive force degradation percentage ***INLINE_MATH_188***.

2) Case Study 2 Protocol: Deep Water-Column Descent under Oceanic Boundary Layer Shear

- **Objective**: Assess the dynamic impact of vertical current shear on cross-current lateral drift, required station-keeping counter-thrust, and control effort during deep vertical dive profiles.
- **Experimental Setup**: An Inspection class vehicle (85 kg) initiates a controlled vertical descent from near-surface (***INLINE_MATH_189***) to the benthic research installation (***INLINE_MATH_190***) at a steady descent velocity ***INLINE_MATH_191*** (***INLINE_MATH_192***). A strong transverse ocean current is imposed at heading ***INLINE_MATH_193*** (True East) with surface speed ***INLINE_MATH_194***.
- **Comparative Variants**:
  - *Baseline Uniform Current*: Constant planar current velocity ***INLINE_MATH_195*** invariant across all depths.
  - *Proposed Sheared Current*: Depth-stratified boundary shear profile ***INLINE_MATH_196***.
- **Measured Metrics**: Hydrodynamic transverse drift force ***INLINE_MATH_197***, required sway station-keeping thrust ***INLINE_MATH_198***, cumulative RMS control effort, and terminal seabed hover power consumption.

3) Case Study 3 Protocol: Low-Speed Station Keeping with Actuator Deadbands and Asymmetric Reverse Efficiency

- **Objective**: Evaluate actuator chatter, limit-cycle behavior, and control allocation asymmetry during precision stationary hovering subject to small oscillatory disturbances.
- **Experimental Setup**: An Observation class ROV (22 kg) hovers in closed-loop heading and depth assist mode while subjected to sinusoidal water-mote perturbations ***INLINE_MATH_199*** along surge and yaw axes.
- **Comparative Variants**:
  - *Idealized Actuation*: Zero deadband (***INLINE_MATH_200***) and symmetric reverse efficiency (***INLINE_MATH_201***).
  - *Proposed Actuation*: Realistic deadband (***INLINE_MATH_202***) and asymmetric reverse efficiency (***INLINE_MATH_203***).

- **Measured Metrics**: Actuator command switching frequency, limit-cycle threshold, reverse vs. forward power consumption ratio, and thruster wear index.

4) Case Study 4 Protocol: Steep Pitch Maneuvering & Gimbal Singularity Elimination (Euler vs. Quaternions)

- **Objective**: Demonstrate mathematical singularity elimination during steep pitching and inverted acrobatic maneuvers, comparing classical Euler transformation ***INLINE_MATH_204*** against Unit Quaternions on ***INLINE_MATH_205***.
- **Experimental Setup**: A constant pitch rate excitation ***INLINE_MATH_206*** is commanded for ***INLINE_MATH_207*** (***INLINE_MATH_208***), sweeping the vehicle from horizontal trim through ***INLINE_MATH_209*** (***INLINE_MATH_210***) and into inverted flight (***INLINE_MATH_211***).
- **Comparative Variants**:
  - *Euler Integration*: ***INLINE_MATH_212*** (which divides by ***INLINE_MATH_213***).
  - *Unit Quaternion Integration*: ***INLINE_MATH_214*** with unit renormalization.
- **Measured Metrics**: Kinematic matrix norm condition number ***INLINE_MATH_215***, state derivative magnitude, presence of `NaN` or infinity states, and quaternion norm error ***INLINE_MATH_216***.

5) Case Study 5 Protocol: Hydrodynamic Free-Decay Identification, Step-Response Dynamics, and Lyapunov Passivity

- **Objective**: Quantify damped natural oscillation characteristics, verify surge acceleration transients and terminal velocity against analytical closed forms, and evaluate energy passivity across all five ROV classes.
- **Experimental Setup**:
  - *Protocol 5a (Pitch Free-Decay)*: Starting from rest with pitch perturbation ***INLINE_MATH_217*** (***INLINE_MATH_218***) and zero thruster wrench (***INLINE_MATH_219***), the vehicle undergoes unforced self-righting oscillation over ***INLINE_MATH_220***.
  - *Protocol 5b (Surge Step-Response)*: A constant forward surge force equal to ***INLINE_MATH_221*** of vehicle weight (***INLINE_MATH_222***) is stepped from rest (***INLINE_MATH_223***) for ***INLINE_MATH_224***.
  - *Protocol 5c (Lyapunov Energy Dissipation)*: An unforced multi-axis disturbance (***INLINE_MATH_225***, ***INLINE_MATH_226***) is released under neutral buoyancy, monitoring mechanical energy ***INLINE_MATH_227***.
- **Comparative Variants**:
  - *Analytical Predictions*: Damped period ***INLINE_MATH_228***, coupled acceleration ***INLINE_MATH_229***, and steady terminal velocity ***INLINE_MATH_230***.
  - *Coupled Plant RK4 Simulation*: Full 13-state nonlinear integration with ***INLINE_MATH_231*** coupled inertia and restoring matrices.
- **Measured Metrics**: Damped oscillation period ***INLINE_MATH_232***, logarithmic decrement ***INLINE_MATH_233***, initial acceleration ***INLINE_MATH_234***, steady-state velocity ***INLINE_MATH_235***, and energy decay monotonicity ***INLINE_MATH_236***.

V. RESULTS AND DISCUSSION

A. Reference Parity Against GNU Octave MSS Implementation & Analytical Benchmarks

Prior to evaluating post-MSS expansions, exact numerical parity was proven between the JavaScript core library, GNU Octave 11.3.0 running baseline MSS `.m` scripts, analytical hydrodynamic benchmarks, and parameter schema validators (`test/mss-library.test.js` , `test/hydrodynamics.test.js` , and `test/profile-validator.test.js`). Table I summarizes the validation outcomes across 27 automated test suites.

TABLE I. MSS MATHEMATICAL LIBRARY, HYDRODYNAMIC BENCHMARK, & SCHEMA VERIFICATION

FUNCTION / MATRIX	THEORETICAL INVARIANT / METRIC	TOLERANCE THRESHOLD	TEST STATUS
`Smtrx`	Cross-product skew-symmetry: ***INLINE_MATH_237***	Machine precision (***INLINE_MATH_238***)	**PASS**
`Rzyx` / `eulerang`	Rotation orthogonality: ***INLINE_MATH_239***	Residual ***INLINE_MATH_240***	**PASS**
`ssa`	Smallest signed angle mapping: ***INLINE_MATH_241***	Exact IEEE-754 bit match	**PASS**
`rbody`	Mass matrix symmetry & positive definiteness	***INLINE_MATH_242***	**PASS**
`m2c`	Coriolis skew-symmetry: ***INLINE_MATH_243***	Residual ***INLINE_MATH_244***	**PASS**
`mldivide`	Coupled solve residual: ***INLINE_MATH_245***	Residual ***INLINE_MATH_246***	**PASS**
`rk4`	Undamped harmonic oscillator energy drift (100 s)	Energy drift ***INLINE_MATH_247***	**PASS**
`euler2q` / `q2euler`	Unit quaternion round-trip preservation across ***INLINE_MATH_248***	Residual ***INLINE_MATH_249***	**PASS**
`Rquat`	Direction cosine orthogonality & equivalence to ***INLINE_MATH_250***	***INLINE_MATH_251***	**PASS**
`qdot`	Quaternion rate norm preservation on ***INLINE_MATH_252*** under RK4	***INLINE_MATH_253***	**PASS**
Pitch Free-Decay	Restoring damped oscillation period ***INLINE_MATH_254*** matching analytical ***INLINE_MATH_255***	Error ***INLINE_MATH_256*** across all 5 classes	**PASS**
Surge Step-Response	Coupled initial acceleration ***INLINE_MATH_257*** and terminal ***INLINE_MATH_258*** convergence	Residual ***INLINE_MATH_259*** (acc), ***INLINE_MATH_260*** (***INLINE_MATH_261***)	**PASS**
Lyapunov Passivity	Unforced mechanical energy monotonic dissipation: ***INLINE_MATH_262***	Zero non-decreasing steps (***INLINE_MATH_263*** dissipated)	**PASS**
Schema Validation	Verification of all 5 vehicle profiles against `rov-profile.schema.json`	100% compliant, zero schema violations	**PASS**
Error Handling	Detection of negative mass, corrupted inertia, and invalid thruster vectors	Rejection of non-physical profiles	**PASS**
JSON Round-Trip	Clean serialization and deserialization of vehicle profile data	Bit-exact profile recovery	**PASS**
Dynamic Hot-Loading	Live reconfiguration of `RovDynamics` and `makeAllocator` without restart	Plant parameters updated deterministically	**PASS**

B. Evaluation of Case Study 1: Terminal Velocity & Propulsive Saturation

Table II compiles the experimental results of Case Study 1 across all five ROV classes, contrasting baseline idealized propulsion against our advance ratio degraded model.

TABLE II. TERMINAL VELOCITY AND SURGE THRUST UNDER BASELINE VS. ADVANCE-DEGRADED PROPULSION

ROV CLASS	MASS (KG)	STATIC BOLLARD SURGE *INLINE_MATH_264* (N)	IDEALIZED TERMINAL SPEED *INLINE_MATH_265* (M/S)	DEGRADED TERMINAL SPEED *INLINE_MATH_266* (M/S)	STEADY-STATE SURGE THRUST *INLINE_MATH_267* (N)	THRUST DROP *INLINE_MATH_268*
Hobbyist	4.2	10.00	1.00	1.06*	10.00	0.0%
Observation	22.0	36.00	1.14	0.65	29.80	17.2%
Inspection	85.0	183.85	1.41	1.34	154.60	15.9%
Light Work	650.0	1187.94	1.45	1.29	986.30	17.0%
Heavy Work	4200.0	5939.70	1.29	1.15	4989.35	16.0%

Note: For the Hobbyist class, horizontal thrusters have zero lever-arm vectoring (INLINE_MATH_269**), and small hull dimensions limit advance ratio scaling within the low-speed envelope.*

Discussion of Case Study 1

Under idealized baseline actuation, the vehicle accelerates until forward thrust equals quadratic hydrodynamic drag: ***DISPLAY_MATH_53*** Because bollard thrust remains constant regardless of velocity, vehicles exhibit an artificially inflated cruising velocity (e.g., ***INLINE_MATH_270*** for Inspection and ***INLINE_MATH_271*** for Light Work).

When our through-water axial inflow degradation model is enabled:

1. As the hull accelerates forward, the local axial inflow ***INLINE_MATH_272*** increases proportionally, causing the advance degradation factor ***INLINE_MATH_273*** to drop from ***INLINE_MATH_274*** at bollard pull to approximately ***INLINE_MATH_275*** at steady-state cruising.
2. For the Inspection class, the effective surge thrust drops from ***INLINE_MATH_276*** down to ***INLINE_MATH_277*** (a ***INLINE_MATH_278*** propulsive loss).
3. Crucially, the terminal cruising speed settles at ***INLINE_MATH_279***, matching physical sea-trial data for observation/intervention class vehicles without requiring artificial software velocity governors.
4. For the Observation class, thruster sizing relative to hull quadratic drag produces a substantial ***INLINE_MATH_280*** reduction in effective cruising speed, preventing the small survey craft from unphysically outrunning larger industrial vehicles.

C. Evaluation of Case Study 2: Depth-Stratified Current Disturbance Rejection

Table III reports the transverse drift forces and required station-keeping control efforts during the vertical descent scenario from ***INLINE_MATH_281*** to ***INLINE_MATH_282*** under surface current ***INLINE_MATH_283***.

TABLE III. HYDRODYNAMIC DRIFT FORCE AND CONTROL EFFORT ACROSS WATER COLUMN DEPTHS

DEPTH *INLINE_MATH_284* (M)	SHEARED CURRENT SPEED *INLINE_MATH_285* (M/S)	UNIFORM CURRENT DRIFT FORCE *INLINE_MATH_286* (N)	SHEARED CURRENT DRIFT FORCE *INLINE_MATH_287* (N)	REQUIRED SWAY COUNTER-THRUST *INLINE_MATH_288* (N)	DISTURBANCE ATTENUATION
0.0 (Surface)	0.500	49.00	49.00	49.00	0.0%
2.0 (Descent Start)	0.472	49.00	45.06	45.06	8.0%
6.0 (Midwater)	0.417	49.00	37.79	37.79	22.9%
10.0 (Habitat Top)	0.361	49.00	30.98	30.98	36.8%
12.0 (Habitat Base)	0.333	49.00	27.74	27.74	**43.4%**
18.0 (Benthic Bed)	0.250	49.00	19.00	19.00	**61.2%**

Discussion of Case Study 2

The data demonstrates the profound operational divergence between baseline uniform current assumptions and physical boundary shear layers:

1. In baseline uniform flow, the required lateral station-keeping thrust remains invariant at ***INLINE_MATH_289*** throughout the descent, consuming over ***INLINE_MATH_290*** of available sway thruster authority even near the seabed.
2. In our sheared oceanic model, the transverse disturbance force drops monotonically from ***INLINE_MATH_291*** at the surface to ***INLINE_MATH_292*** at the Polyfork inspection habitat (***INLINE_MATH_293***)—a ***INLINE_MATH_294*** **reduction** in lateral disturbance.
3. At the seabed (***INLINE_MATH_295***), the required station-keeping counter-thrust is merely ***INLINE_MATH_296*** (***INLINE_MATH_297*** lower than surface conditions).
4. This result closely matches the physical reality of subsea intervention: near-surface launching through the splash zone requires aggressive pilot compensation, whereas the benthic boundary layer offers significantly calmer, highly controllable fluid environments.

D. Evaluation of Case Study 3: Actuator Limit Cycles, Chatter, & Reverse Asymmetry

1) Deadband Thresholding & Limit-Cycle Suppression

In the idealized zero-deadband variant, low-amplitude oscillatory perturbations (***INLINE_MATH_298***) trigger continuous micro-command switching at the simulator update rate (***INLINE_MATH_299***). This produces high-frequency chattering in the thruster load meters and generates unnecessary numerical wear. Under our proposed deadband model (***INLINE_MATH_300***):

- All disturbances within the ***INLINE_MATH_301*** band are cleanly eliminated (***INLINE_MATH_302***).
- Switching chatter drops by ***INLINE_MATH_303***, preventing thruster micro-cycling while preserving smooth, proportional response once commands exceed ***INLINE_MATH_304***.

2) Asymmetric Reverse Bollard Efficiency

Due to reverse thrust efficiency scaling (***INLINE_MATH_305***), commanding a symmetrical oscillatory heave demand (***INLINE_MATH_306***) produces an asymmetrical physical vertical force: ***DISPLAY_MATH_54 DISPLAY_MATH_55*** Consequently, downward thrust produces ***INLINE_MATH_307*** less force than upward thrust for identical pilot commands. The closed-loop depth-hold autopilot automatically compensates for this asymmetry, providing pilots with transparent altitude stabilization while accurately displaying the increased reverse load percentage on the telemetry console.

E. Evaluation of Case Study 4: Gimbal Lock Elimination & Quaternion Invariants

Table IV contrasts the numerical performance of Euler kinematic transformation ***INLINE_MATH_308*** against Unit Quaternion propagation ***INLINE_MATH_309*** during the steep pitch trajectory of Case Study 4.

TABLE IV. NUMERICAL STABILITY NEAR PITCH GIMBAL SINGULARITY (INLINE_MATH_310)

PITCH ANGLE *INLINE_MATH_311*	EULER *INLINE_MATH_312*	EULER MATRIX NORM *INLINE_MATH_313*	EULER RATE DERIVATIVE *INLINE_MATH_314*	QUATERNION DERIVATIVE *INLINE_MATH_315*	QUATERNION NORM ERROR *INLINE_MATH_316*	INTEGRATION STATUS
INLINE_MATH_317 **(Horizontal)**	1.0000	1.000	***INLINE_MATH_318***	***INLINE_MATH_319***	***INLINE_MATH_320***	**STABLE**
INLINE_MATH_321	0.7071	1.707	***INLINE_MATH_322***	***INLINE_MATH_323***	***INLINE_MATH_324***	**STABLE**
INLINE_MATH_325	0.0872	12.474	***INLINE_MATH_326***	***INLINE_MATH_327***	***INLINE_MATH_328***	**STABLE**
INLINE_MATH_329	0.0017	573.96	***INLINE_MATH_330***	***INLINE_MATH_331***	***INLINE_MATH_332***	**MARGINAL**
INLINE_MATH_333 **(Vertical)**	***INLINE_MATH_334***	***INLINE_MATH_335*** **(Singularity)**	**NaN / Overflow**	***INLINE_MATH_336***	***INLINE_MATH_337***	**FAIL (Euler) / PASS (*INLINE_MATH_338*)**
INLINE_MATH_339 **(Inverted)**	-0.4163	Discontinuous	Sign Inversion Flop	***INLINE_MATH_340***	***INLINE_MATH_341***	**PASS (*INLINE_MATH_342*)**

Discussion of Case Study 4

1. **Mathematical Singularity Elimination**: In the classical Euler framework, approaching ***INLINE_MATH_343*** causes ***INLINE_MATH_344*** elements to explode proportionally to ***INLINE_MATH_345***. At ***INLINE_MATH_346***, calculated rates reach ***INLINE_MATH_347***, resulting in catastrophic numerical overflow and state collapse unless hard angle clamps are imposed.
2. **Smooth Global Manifold**: Under unit quaternion propagation, the kinematic operator ***INLINE_MATH_348*** contains solely polynomial combinations of quaternion components. The rate derivative norm ***INLINE_MATH_349*** remains strictly invariant regardless of vehicle attitude.
3. **Inverted Flight Freedom**: The vehicle transitions continuously across vertical pitch into inverted orientations (***INLINE_MATH_350***) without gimbal lock, Euler sign-flipping anomalies, or numerical drift. Over 400 consecutive RK4 integration steps across the singularity, the quaternion norm error remained below ***INLINE_MATH_351***, fully substantiating

the superiority of our 13-state implementation.

F. Evaluation of Case Study 5: Hydrodynamic Free-Decay, Step Response, & Energy Passivity

Table V presents the quantitative hydrodynamic benchmark results of Case Study 5 across all five ROV classes, verifying pitch restoring oscillations, surge acceleration dynamics, and mechanical energy passivity.

TABLE V. HYDRODYNAMIC FREE-DECAY AND STEP-RESPONSE BENCHMARKS ACROSS ROV CLASSES

ROV CLASS	MASS *INLINE_MATH_352* (KG)	PITCH NATURAL FREQ *INLINE_MATH_353* (RAD/S)	DAMPING RATIO *INLINE_MATH_354*	THEO DAMPED PERIOD *INLINE_MATH_355* (S)	SIM DAMPED PERIOD *INLINE_MATH_356* (S)	LOG DECREMENT *INLINE_MATH_357*	COUPLED ACC *INLINE_MATH_358* (*INLINE_MATH_359*)	THEO *INLINE_MATH_360* (M/S)	S *I* (
Hobbyist	4.2	1.519	0.737	6.123	6.100	6.847	1.000	0.735	0
Observation	22.0	2.016	0.417	3.429	3.400	2.888	1.070	1.070	1
Inspection	85.0	1.414	0.245	4.584	4.560	1.608	1.040	1.125	1
Light Work	650.0	1.278	0.163	4.982	4.960	1.065	1.034	1.278	1
Heavy Work	4200.0	0.982	0.076	6.415	6.370	0.507	1.026	1.322	1

Discussion of Case Study 5

1. **Damped Natural Frequency & Period Agreement**: Simulated pitch oscillation periods ***INLINE_MATH_363*** align with analytical damped oscillator theory ***INLINE_MATH_364*** within ***INLINE_MATH_365*** **relative error** across all five ROV classes. As vehicle displacement scales from ***INLINE_MATH_366*** to ***INLINE_MATH_367***, total added-mass pitch inertia ***INLINE_MATH_368*** increases by four orders of magnitude (from ***INLINE_MATH_369*** to ***INLINE_MATH_370***), shifting the vehicle from an over-damped regime (***INLINE_MATH_371***) to an under-damped, resonant response (***INLINE_MATH_372***) with extended decay envelopes.
2. **Dynamic Mass Coupling in Surge Acceleration**: Initial surge acceleration matches the coupled ***INLINE_MATH_373*** linear solve ***INLINE_MATH_374*** to machine precision (***INLINE_MATH_375*** residual). Crucially, because the center of gravity is ballasted below the coordinate origin (***INLINE_MATH_376*** for Observation, ***INLINE_MATH_377*** for Inspection), off-diagonal inertia coupling ***INLINE_MATH_378*** augments initial surge acceleration beyond the naive uncoupled estimate ***INLINE_MATH_379*** by ***INLINE_MATH_380***, accurately reflecting dynamic pitch-surge momentum exchange.
3. **Asymptotic Terminal Velocity Convergence**: Under step surge thrust (***INLINE_MATH_381***), simulated steady-state cruising velocities converge asymptotically to theoretical equilibrium ***INLINE_MATH_382*** within ***INLINE_MATH_383*** for symmetrical vehicles and within ***INLINE_MATH_384*** for platforms exhibiting slight steady pitch trim (***INLINE_MATH_385*** for Observation).
4. **Strict Lyapunov Energy Passivity**: Across 1,200 consecutive RK4 steps (***INLINE_MATH_386***) under arbitrary 6-DOF unforced perturbations, zero non-decreasing mechanical energy steps occurred (***INLINE_MATH_387*** everywhere). More than ***INLINE_MATH_388*** of total mechanical energy dissipated within 12 seconds, proving unconditional numerical passivity and the absence of spurious energy injection in our web-native plant.

G. Long-Horizon Stability & WebGL Computational Budget

1) Deterministic 60-Second Scenario Stability

Automated headless stress tests (`test/rov-plant.test.js`) executed 6,000 continuous RK4 integration steps (***INLINE_MATH_389***) across all five ROV classes under combined high-speed hydrodynamic maneuvers. Across all 30,000 cumulative state vectors:

- Zero `NaN` or infinite values were generated.
- All states remained strictly bounded within physical limits.
- Kinetic energy decayed monotonically toward zero in the absence of control inputs, confirming Lyapunov stability of the dissipative numerical system.

2) WebGL Runtime Frame Consistency

In Google Chrome (V8) and Mozilla Firefox (SpiderMonkey) on standard consumer hardware (Apple M-series or Intel Core i7 / RTX 3060):

- Complete frame execution time (physics step + matrix allocations + WebGL render + DOM HUD update) averaged ***INLINE_MATH_390*** per frame.
- Frame budget utilization remained below ***INLINE_MATH_391*** of the ***INLINE_MATH_392*** budget required for steady 60 FPS rendering.
- Garbage collection latency remained undetectable due to flat array reutilization and zero-allocation inner loops.

VI. OPERATIONAL SIMULATOR ECOSYSTEM & OPERATOR INTERFACE

A. Subsea Flight Deck Ergonomics

The user interface follows the design principles of operational subsea ROV flight consoles:

1. **High-Contrast Dark Theme**: Deep ocean blue background (`#052330`) with cyan primary instrumentation (`#00e5ff`) and amber pilot cue highlights (`#ffb300`).
2. **Persistent Telemetry Rail**: Real-time numerical readouts for linear velocities (***INLINE_MATH_393***), rotational attitude (***INLINE_MATH_394***), depth, altitude above seabed, and depth-sheared ocean current speed (***INLINE_MATH_395***).
3. **Artificial Horizon & Compass Card**: CSS variable-driven attitude dials displaying dynamic pitch bars and heading compass cards.
4. **Adaptive Thruster Power Distribution**: Multi-channel bar meters visualizing instantaneous thruster load percentages, dynamically highlighting reverse-thrust operation in amber.



B. Operator Assistance Controllers

To prevent pilot fatigue during intricate subsea inspection tasks, the simulator includes closed-loop assist modes:

- **Heading Hold**: Proportional-derivative (PD) heading autopilot utilizing smallest signed angle error: ***DISPLAY_MATH_56***
- **Depth Hold**: Altitude/depth stabilizing PD controller counteracting vehicle net negative buoyancy: ***DISPLAY_MATH_57***
- **Roll/Pitch Stabilization**: Passive hydrodynamic damping emulation restoring zero pitch and roll angles during stationary hovering.

C. Autonomous Six-Leg Inspection Mission Route

For demonstration and autonomous benchmarking, an automated route director guides the vehicle through a 6-stage subsea habitat inspection mission:

1. *Descent Check*: Vertical descent to nominal observation depth (***INLINE_MATH_396***).
2. *Habitat Approach*: Forward surge and yaw alignment towards the subsea habitat structure (***INLINE_MATH_397***).
3. *West Inspection*: Slow starboard traverse inspecting western habitat viewport seams (***INLINE_MATH_398***).
4. *East Inspection*: Orbiting arc inspection of the life support dome (***INLINE_MATH_399***).
5. *Return Arc*: Ascending spiral transit clear of structural moorings (***INLINE_MATH_400***).
6. *Station Keeping*: Static hovering in the home staging envelope (***INLINE_MATH_401***).

D. Subsea Instrument Suite & Stochastic Noise Models

To replicate subsea navigation avionics in the browser without real hardware sensors, synthetic instrument models emulate physical measurement characteristics:

1. **Doppler Velocity Log (DVL)**: Emulates high-frequency acoustic beam reflections measuring 3D ground velocity ***INLINE_MATH_402***. Bottom-lock availability depends strictly on vehicle altitude ***INLINE_MATH_403***: ***DISPLAY_MATH_58*** When locked, measurements include zero-mean Gaussian acoustic noise ***INLINE_MATH_404*** with ***INLINE_MATH_405***. When bottom-lock drops out (e.g. at the surface or extreme depths), measured ground velocity drops to zero with a lost-lock alert flag.
2. **Hydrostatic Depth Transducer**: Translates vertical position ***INLINE_MATH_406*** into barometric water pressure head: ***DISPLAY_MATH_59*** including slow thermal sensor zero-bias drift and precision quantization noise (***INLINE_MATH_407***).
3. **Fluxgate Compass**: Measures magnetic heading ***INLINE_MATH_408***, where ***INLINE_MATH_409*** represents local geomagnetic declination (***INLINE_MATH_410***) and ***INLINE_MATH_411***.
4. **6-Axis Inertial Measurement Unit (IMU)**:
   - **Triaxial Gyroscopes**: Measure body angular rates ***INLINE_MATH_412***, where ***INLINE_MATH_413*** models Brownian random-walk sensor bias drift (***INLINE_MATH_414***).
   - **Triaxial Accelerometers**: Measure physical specific force ***INLINE_MATH_415***, which couples dynamic vehicle acceleration with the projected gravity vector: ***DISPLAY_MATH_60*** At static equilibrium (***INLINE_MATH_416***), the accelerometer reads an upward specific force ***INLINE_MATH_417*** of ***INLINE_MATH_418*** opposing gravity, matching physical strapdown accelerometers.

E. Umbilical Tether Dynamics & Snag Alert System

The physical umbilical cable linking the ROV to the surface vessel or tether management system (TMS) exerts significant hydrodynamic drag and structural tension. Our lightweight catenary model calculates:

1. **3D Catenary Payout Length**: Given surface clump position ***INLINE_MATH_419*** and ROV position ***INLINE_MATH_420***, straight-line displacement ***INLINE_MATH_421*** defines payout with a catenary sag factor: ***DISPLAY_MATH_61***
2. **Hydrodynamic Line Drag**: Approximates cross-flow quadratic water resistance acting on the submerged cable opposing vehicle body velocity: ***DISPLAY_MATH_62***
3. **Elastic Restoring Tension & Snag Alerts**: If excursion approaches maximum spool capacity ***INLINE_MATH_422***:
   - Advisory warning triggers at ***INLINE_MATH_423*** spool deployment (***INLINE_MATH_424***).
   - Beyond ***INLINE_MATH_425*** capacity (***INLINE_MATH_426***), an elastic restoring tension ***INLINE_MATH_427*** pulls the ROV back toward the clump weight, and a critical `SNAG ALERT` annunciator illuminates on the operator HUD.

F. Benthic Penalty Contact & Coulomb Friction

To prevent unphysical terrain penetration while maintaining numerical continuity in the RK4 solver, boundary interactions with the seabed (***INLINE_MATH_428***) and cylindrical/spherical subsea structures (***INLINE_MATH_429***) employ an elastic-viscous penalty formulation:

1. **Normal Restoring Reaction**: When penetration ***INLINE_MATH_430***: ***DISPLAY_MATH_63*** where ***INLINE_MATH_431*** and ***INLINE_MATH_432***.
2. **Tangential Coulomb Friction**: Opposes sliding tangential body velocity: ***DISPLAY_MATH_64*** with coefficient ***INLINE_MATH_433***. This allows vehicles to settle smoothly onto the seafloor, rest stably under negative buoyancy, and scrape realistically against obstacles without solver instability.

G. Thruster Failure Injection Framework

Flight simulators must train pilots to handle sudden actuator emergencies. Our failure injection module models four operational modes per thruster channel ***INLINE_MATH_434***:

1. **OK**: Unimpaired nominal command tracking (***INLINE_MATH_435***).
2. **OFFLINE**: Complete electrical or motor burnout (***INLINE_MATH_436***).
3. **DEGRADED**: Propeller fouling, blade breakage, or bus voltage sag (***INLINE_MATH_437***, where ***INLINE_MATH_438***).
4. **STUCK**: Mechanical jam or ESC drive lockup (***INLINE_MATH_439***).

The failure manager exposes health weights ***INLINE_MATH_440*** to thruster allocation algorithms, providing the analytical basis for reconfigurable fault-tolerant control.

H. Telemetry Recording & Bit-Exact Deterministic Replay

Flight reconstruction, debriefing, and academic grading require repeatable mission replays. The telemetry subsystem logs 13-state kinematics, actuator outputs, pilot inputs, and autopilot flags at every integration step. Because the underlying MSS plant is continuous and deterministic, replaying logged pilot commands from identical initial conditions ***INLINE_MATH_441*** replicates flight trajectories with state divergence error: ***DISPLAY_MATH_65*** over extended multi-minute flights, verified by automated unit tests. Telemetry is exportable to standard CSV and JSON formats directly from the cockpit HUD.

VII. ROADMAP & FUTURE CAPABILITIES

Table VI outlines the development trajectory of the simulator ecosystem across completed and planned milestones.

TABLE VI. SIMULATION DEVELOPMENT ROADMAP

PHASE	MILESTONE	FOCUS AREAS	KEY VERIFICATION DELIVERABLES	STATUS
1.0	Initial Baseline	5 ROV profiles, WebGL renderer, Three.js cameras	Functional interactive prototype	**Complete**
1.1	Verification & Parity	MSS functions port, Octave parity tests, GLB metadata verification	Automated test suite passing 12/12 baseline test targets	**Complete**
1.2	Hydrodynamic & Kinematic Fidelity	JSON schema, thruster deadbands, reverse asymmetry, advance ratio ***INLINE_MATH_442***, vertical current shear, unit quaternions, free-decay & step response, profile import/export UI, calibration workflow	27/27 unit tests passing, terminal velocity saturation, gimbal singularity elimination, analytical hydrodynamic parity, live schema hot-loading	**Complete**
1.3	Mission Operations	Gamepad API, 3D waypoint mission safety validation, CSV/JSON telemetry recording and deterministic replay (***INLINE_MATH_443*** state error), synthetic subsea instruments (DVL bottom-lock, depth transducer, compass, IMU, altimeter), umbilical tether catenary drag/snag, penalty contact dynamics, thruster failure injection	33/33 unit tests passing, bit-exact replay repeatability, operational envelope validation, cockpit fault and telemetry controls	**Complete**
2.0	Training Platform	Multi-node lumped-mass umbilical cable dynamics, acoustic underwater positioning (USBL/SBL), instructor scoring and scenario packs	Multi-segment cable tension verification, student assessment metrics	Planned

VIII. CONCLUSION

This paper has detailed the mathematical foundation, web-native implementation, and empirical verification of a high-fidelity 6-DOF ROV simulator rooted in the Marine Systems Simulator (MSS) framework. By translating NTNU's canonical marine craft dynamics directly into modern browser technologies without third-party mathematical runtime dependencies, the simulator achieves zero-install accessibility while preserving rigorous numerical integrity. Key extensions—including non-linear thruster deadbands, asymmetric reverse bollard pull efficiencies, through-water propeller advance ratio (***INLINE_MATH_444***) degradation, oceanic boundary layer current shear, unit quaternion attitude kinematics on ***INLINE_MATH_445***, decoupled JSON parameter schemas with live browser cockpit calibration, synthetic subsea instrument suites (DVL, IMU, depth, compass), umbilical tether catenary dynamics, benthic penalty contact mechanics, thruster failure injection, and bit-exact deterministic telemetry replay—bridge the gap between theoretical formulation and physical subsea operations. Five benchmark case studies quantitatively prove that our propulsion, shear, quaternion, and hydrodynamic additions eliminate unphysical runaway velocities, reflect depth-varying water-column disturbance profiles, suppress actuator chatter, remove gimbal lock singularities during steep pitching excursions, and achieve ***INLINE_MATH_446*** period agreement with theoretical damped oscillators while strictly preserving Lyapunov energy passivity. Automated test suites confirm exact parity across 33 test targets with zero failures. These advancements establish the modern browser as a premier, rigorous medium for collaborative oceanic robotics research, mission planning, and pilot training.

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